High-dose chemotherapy followed by autologous stem-cell transplantation versus non-myeloablative consolidation in primary CNS lymphoma (MATRix/IELSG43): a randomised phase 3 trial

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Summary

Background Consolidation therapy for primary CNS lymphoma (PCNSL) includes high-dose chemotherapy with autologous stem-cell transplantation (HCT–ASCT), yet its efficacy compared with non-myeloablative chemoimmunotherapy remains uncertain. This study aimed to provide randomised comparative evidence on the efficacy and safety of thiotepa-based HCT–ASCT versus non-myeloablative R-DeVIC consolidation after uniform MATRix induction in newly diagnosed PCNSL.

Methods This open-label, randomised, phase 3 trial was conducted across 56 university and academic non-university hospitals with established transplantation facilities in five European countries. Eligible for inclusion were untreated, immunocompetent patients with B-cell PCNSL, aged 18–65 years regardless of Eastern Cooperative Oncology Group (ECOG) performance status, or 66–70 years with ECOG performance status 0–2. Pretreatment corticosteroids were permitted. Exclusion criteria included lymphoma manifestation outside the CNS, and congenital or acquired immunodeficiency. Patients received four cycles of MATRix induction (comprising rituximab, high-dose cytarabine, and thiotepa, in addition to high-dose methotrexate). Patients reaching at least partial response were randomly assigned (1:1) to two cycles of R-DeVIC (rituximab plus dexamethasone, etoposide, ifosfamide, and carboplatin) or HCT–ASCT with carmustine and thiotepa. The primary endpoint was progression-free survival, assessed in the full analysis set (excluding patients with major violations of entry criteria). The safety analysis set contained all randomised patients who initiated therapy. This study is registered with ClinicalTrials.gov (NCT02531841) and the EU Clinical Trials Register (EudraCT number 2012-000620-17). The study is completed.

Findings Between July 16, 2014, and Aug 31, 2019, 368 patients were enrolled. 346 (94%) patients started induction treatment, and 230 were randomly assigned; 229 patients were analysed (R-DeVIC group, n=115; HCT–ASCT group, n=114). After a median follow-up of 45·3 months, 3-year progression-free survival was significantly superior in the HCT–ASCT group (hazard ratio 0·43 [95% CI 0·27–0·68]; p=0·0003). The 3-year progression-free survival was 78% (95% CI 69–85) in the HCT–ASCT group compared with 51% (41–60) in the R-DeVIC group. The mean number of adverse events per patient was 9·3 (SD 4·4) in the R-DeVIC group and 14·6 (5·8) in the HCT–ASCT group. Fatal serious adverse events following consolidation treatment occurred in two patients in the R-DeVIC group (both acute myeloid leukaemia) and in five patients in the HCT–ASCT group (infections and infestations [n=4], pulmonary embolism [n=1]); all of these events apart from the pulmonary embolism were judged to be possibly related to treatment.

Interpretation In the largest randomised trial in untreated PCNSL to date, HCT–ASCT significantly improved progression-free and overall survival compared with non-myeloablative consolidation in patients who had completed induction treatment, establishing it as the preferred consolidation strategy in fit patients.

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Research in context

Evidence before this study

Before designing this trial, we searched PubMed for studies published from Jan 1, 1990, to Dec 31, 2013, using the terms "central nervous system", "CNS", "lymphoma", "primary CNS lymphoma", "PCNSL", "autologous stem cell transplantation", "high-dose chemotherapy", "DeVIC", "consolidation", and "first-line treatment". We also screened reference lists of relevant articles, conference proceedings from the American Society of Hematology, American Society of Clinical Oncology, European Hematology Association, International Conference on Malignant Lymphoma, and trial registries for ongoing or unpublished studies. We identified ten—mostly small, single-arm phase 2—studies showing that high-dose chemotherapy followed by autologous stem-cell transplantation (HCT–ASCT) was feasible in newly diagnosed primary CNS lymphoma (PCNSL). A prospective phase 2 study in 79 patients further supported the activity of thiotepa-based HCT–ASCT in the first-line setting, with encouraging long-term disease control at manageable toxicity. Since this trial was initiated, larger prospective single-arm studies and long-term follow-up cohorts have confirmed sustained efficacy of thiotepa-based HCT–ASCT after high-dose methotrexate-based induction, although with relevant acute toxicity and non-negligible treatment-related mortality. Two randomised phase 2 trials, IELSG32 and PRECIS, compared HCT–ASCT with whole-brain radiotherapy as consolidation and supported HCT–ASCT as an effective alternative with less risk of delayed neurotoxicity. Non-myeloablative chemotherapy has also been explored as an alternative consolidation strategy.

In the Alliance/CALGB 51101 randomised phase 2 trial, patients were randomly assigned before induction to a consolidation strategy consisting of thiotepa-based HCT–ASCT or non-myeloablative chemotherapy with etoposide and cytarabine. However, interpretation is complicated by an imbalance in progressive disease during induction between groups, which might have favoured the transplantation group. No completed randomised phase 3 trial had directly compared these approaches after uniform high-dose methotrexate-based induction in newly diagnosed PCNSL. Thus, the optimal consolidation strategy remained uncertain.

Added value of this study

The MATRix/IELSG43 trial is a multicentre study involving 56 centres across five European countries, with detailed clinical annotation, and the first phase 3 trial to directly compare thiotepa-based HCT–ASCT with non-myeloablative R-DeVIC (rituximab plus dexamethasone, etoposide, ifosfamide, and carboplatin) consolidation after a standardised MATRix induction in patients with newly diagnosed PCNSL. Among patients up to 70 years of age who had at least partial response after MATRix, HCT–ASCT significantly improved 3-year progression-free survival (78% *vs* 51%; hazard ratio [HR] 0·43) and overall survival (86% *vs* 71%; HR 0·46) compared with R-DeVIC, with acceptable toxicity in experienced centres. Quality-of-life data showed numerically improved, global health status over time in the HCT–ASCT group.

Our results provide high-level evidence that confirms and extends findings from earlier phase 2 studies. By using a uniform, widely adopted induction regimen (MATRix) and randomising patients only after response, this trial isolates the effect of consolidation strategy and clarifies that HCT–ASCT is superior to an intensive non-myeloablative chemoimmunotherapy approach in this setting.

Implications of all the available evidence

Taken together with previous trials, our findings support thiotepa-based HCT–ASCT as the preferred consolidation for fit PCNSL patients who respond to intensive high-dose methotrexate-based induction, where transplantation expertise is available. Non-myeloablative multi-agent chemotherapy, such as R-DeVIC, could be a therapeutic option for patients who are not candidates for HCT–ASCT, but should be further evaluated within clinical trials.

At the same time, approximately one-third of patients in MATRix/IELSG43 did not reach randomisation due to induction toxicity or early progression, underscoring that optimisation of induction regimens is urgently needed. Ongoing phase 3 studies exploring modified high-dose methotrexate based induction treatment and trials integrating novel agents such as Bruton's tyrosine kinase inhibitors, immunomodulatory drugs, and chimeric antigen receptor T-cell therapy are likely to redefine treatment strategies, particularly for older or frailer patients. Until such data mature, international guidelines and clinical practice should consider MATRix followed by thiotepa-based HCT–ASCT as a standard first-line approach for eligible patients with newly diagnosed PCNSL.

Introduction

Primary CNS lymphoma (PCNSL) is a rare lymphoid malignancy[1,2] with an aggressive biology leading to considerable morbidity and mortality. Thus, prompt diagnosis and effective therapy are crucial elements for improving cure rates and preserving neurocognitive function. Treatment principles applied to extracerebral systemic large B-cell lymphoma, such as R-CHOP (rituximab, cyclophosphamide, vincristine, prednisone), cannot be transferred to PCNSL as the blood–brain barrier hampers CNS bioavailability of these drugs.

The introduction of high-dose methotrexate-based therapy has improved survival in recent decades. Based on the results of the IELSG32 study, the MATRix regimen,[3] comprising rituximab, high-dose cytarabine, and thiotepa, in addition to high-dose methotrexate, is an established induction regimen with high response rates for young and fit patients and an overall response rate of 79%.[4]

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Subsequent consolidative therapy options include high-dose chemotherapy supported by autologous stem-cell transplantation (HCT–ASCT). The efficacy of thiotepa-based HCT–ASCT after high-dose methotrexate-based induction treatment has been shown for eligible patients irrespective of age; yet, treatment-related toxicity and mortality remain a challenge.[4–9] Despite encouraging results from phase 2 trials, it remains uncertain whether intensive HCT–ASCT offers a net benefit over conventional non-myeloablative consolidation, given its potentially higher risk of treatment-related morbidity and mortality. Direct comparison in a randomised phase 3 trial was therefore necessary to define the optimal consolidation strategy for newly diagnosed PCNSL, balancing efficacy, survival outcomes, and treatment tolerability.

Aiming to reduce treatment-related morbidity, alternative non-myeloablative consolidation strategies have been developed. The non-myeloablative DeVIC regimen (dexamethasone, etoposide, ifosfamide, and carboplatin) has shown an acceptable tolerability and was highly effective when followed by whole-brain radiotherapy in untreated PCNSL patients.[10]

From these data, we proposed that the non-myeloablative DeVIC consolidation strategy, especially when combined with rituximab (R-DeVIC), might result in comparable survival results with lower morbidity compared with HCT–ASCT in the MATRix/IELSG43 trial.

Here, we report the progression-free survival results from the randomised, phase 3 MATRix/IELSG43 trial comparing the non-myeloablative chemotherapy combination DeVIC, in addition to rituximab (R-DeVIC), with ASCT using carmustine and thiotepa conditioning in patients with untreated PCNSL in complete or partial response after four courses of MATRix.

Methods

Study design

MATRix/IELSG43 was a multicentre, open-label, randomised, phase 3 trial with two parallel treatment arms. It was conducted at 56 trial sites in five countries (Germany, Italy, Denmark, Norway, and Switzerland) within the networks of the German Cooperative PCNSL study group and the International Extranodal Lymphoma Study Group (IELSG). The sites were university and academic non-university hospitals with established transplantation facilities. The trial methods, conduct, and analysis are described in the published clinical trial protocol[11] and statistical analysis plan (appendix). The investigators designed and conducted the trial, with support from the Clinical Trials Unit at the University Medical Centre Freiburg. All authors vouch for the completeness and accuracy of the data and for the fidelity of the trial to the protocol. Ethics approval was granted by the Ethics Committee of the Landesärztekammer Baden-Württemberg in Stuttgart, Germany (approval number AM-2014-010-ff) and the ethics committees of the participating centres (appendix p 5). The study conformed to the tenets of the Declaration of Helsinki and institutional board requirements. This study is registered with ClinicalTrials.gov (NCT02531841) and the EU Clinical Trials Register (EudraCT number 2012-000620-17). The study is completed.

Participants

Eligible for the study were untreated, immunocompetent patients with B-cell PCNSL, aged 18–65 years regardless of Eastern Cooperative Oncology Group (ECOG) performance status, or 66–70 years with ECOG performance status 0–2. Pretreatment corticosteroids were permitted. The diagnostic sample, obtained via biopsy, cerebrospinal fluid cytology, or vitrectomy, had to show B-cell lymphoma on local histological or cytological evaluation, with subsequent confirmation by central neuropathological review. Exclusion criteria included lymphoma manifestation outside the CNS, congenital or acquired immunodeficiency, including HIV seropositivity, renal impairment (defined as creatinine clearance <60 mL/min, as per Modification of Diet in Renal Disease estimation), or inadequate cardiac, pulmonary, or hepatic function at the investigator's discretion, as well as active hepatitis B or C virus disease. Patients with isolated ocular lymphoma without manifestation in the brain parenchyma or spinal cord were not enrolled. All patients or their legal representatives, in cases where patients were temporarily not legally competent, provided written informed consent. Data on patient race or ethnicity were not collected as part of the study protocol, as these variables are not routinely captured across participating centres in this multinational European trial. Patients and members of the public were not involved in the design, conduct, or reporting of the study.

Randomisation and masking

Patients reaching at least partial response after completion of induction treatment with sufficient stem-cell harvest ($\geq 3\times10^6$ $CD34^+$ cells per kg of bodyweight) were randomly assigned (1:1) to either the R-DeVIC group or the HCT–ASCT group (ie, immediately before starting the randomised treatment and after re-checking inclusion and exclusion criteria). Randomisation was stratified by response status (complete or partial response), using block randomisation with two randomly varying block sizes not disclosed to the centres. The randomisation code was produced by validated programmes based on SAS software.

Patients not qualifying for the randomisation were further documented and evaluated for off-protocol therapy. This included patients who had disease progression at any time during the study, patients who did not reach at least partial response at the end of induction treatment, patients who had inadequate bone marrow recovery that caused a treatment delay of at least 4 weeks, or whose stem-cell harvest was insufficient after three induction treatment cycles.

The central neuroradiological review of response assessments was conducted blinded to the allocated treatment group, but treatment allocation was not masked to participants, local outcome assessors, investigators, and data analysts.

Procedures

Induction treatment consisted of four cycles of intravenous rituximab 375 mg/m² per day (days 0 and 5), methotrexate 3·5 g/m² (day 1), cytarabine 2 g/m² twice per day (days 2 and 3), and thiotepa 30 mg/m² (day 4), every 21 days (MATRix regimen).[3] Stem cells were harvested after the second cycle.

Treatment in the R-DeVIC group comprised two cycles of the R-DeVIC regimen: intravenous rituximab 375 mg/m² (day 0), dexamethasone 40 mg absolute per day (days 1–3), etoposide 100 mg/m² per day (days 1–3), ifosfamide 1500 mg/m² per day (days 1–3), and carboplatin 300 mg/m² (day 1), every 21 days. Treatment in the HCT–ASCT group consisted of intravenous HCT with carmustine 400 mg/m² (day –6)—or, if carmustine was unavailable, intravenous busulfan 3·2 mg/kg per day (days –8 and –7)—and thiotepa 5 mg/kg twice per day (days –5 and –4) followed by reinfusion of stem cells on day 0.

Response status was assessed by contrast-enhanced MRI after the second and fourth MATRix cycle during induction (response assessment points 1 and 2) and on day 60 following randomisation (response assessment point 3), both for the R-DeVIC group and the HCT–ASCT group. Thereafter, response assessments were performed every 3 months during years 1–2 and every 6 months during years 3–5. The response was evaluated according to the International PCNSL Cooperative Group response criteria[12] by local neuroradiological assessment. Additional central neuroradiological review was performed during the treatment period and thereafter in case of progression and borderline cases. In cases of leptomeningeal or ocular involvement, as assessed by cerebrospinal fluid cytology and/or flow cytometry and ophthalmic slit lamp examination at screening, these examinations were repeated until negative and considered regarding the respective response evaluation. For further details, refer to the clinical trial protocol and statistical analysis plan in the appendix.

Outcomes

The primary endpoint was progression-free survival, defined as the time from randomisation until the date of progressive disease, relapse, or death from any cause. Patients who did not present with the event of interest were treated as censored observations at the date of the last visit. Secondary endpoints were (1) complete response rate on day 60 after randomisation; (2) overall survival, defined as the time from randomisation to date of the last visit (censored observation) or of death from any cause; (3) duration of response, defined as the time from complete response, unconfirmed complete response, or partial response until relapse or progressive disease; (4) quality of life, measured with the European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire (EORTC QLQ-C30); and (5) safety endpoints, namely adverse events, toxicity (according to the National Cancer Institute's Common Terminology Criteria for Adverse Events version 4.0), and neurotoxicity (according to Mini-Mental State Examination, EORTC QLQ-BN20, and a neuropsychological test battery designed for prospective evaluation in PCNSL[13]). A detailed analysis of quality of life and neurotoxicity will be reported elsewhere.

Adverse events were recorded starting at the first administration of study medication until day 60 after

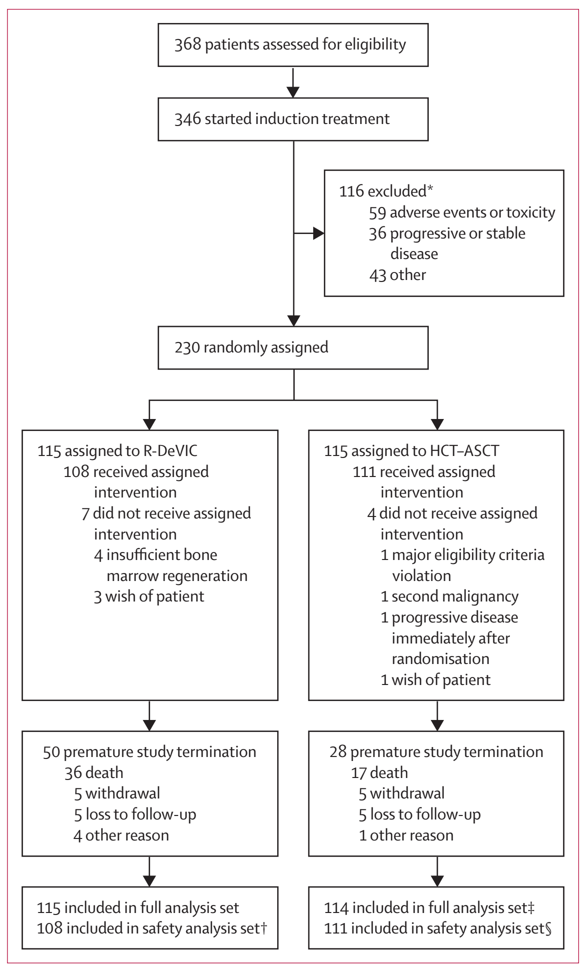

Figure 1: **Trial profile**
R-DeVIC=rituximab plus dexamethasone, etoposide, ifosfamide, and carboplatin. HCT–ASCT=high-dose chemotherapy followed by autologous stem-cell transplantation. *Multiple reasons can apply. Most frequent among other reasons were insufficient stem-cell harvest (n=11), wish of patient (n=7), and screening failure (n=7). †Seven patients who did not start the assigned treatment were excluded. ‡One patient with a major eligibility criteria violation was excluded. §Three patients who did not start the assigned treatment were excluded.

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See Online for appendix

	R-DeVIC group (n=115)	HCT–ASCT group (n=114)
Age, years		
Median (IQR)	60 (54–64)	58 (51–64)
≥65 years	28 (24%)	23 (20%)
<65 years	87 (76%)	91 (80%)
Sex at birth		
Female	53 (46%)	49 (43%)
Male	62 (54%)	65 (57%)
ECOG performance status		
0	18 (16%)	25 (22%)
1	70 (61%)	56 (49%)
2	16 (14%)	29 (25%)
3	7 (6%)	3 (3%)
4	3 (3%)	0
Missing	1 (1%)	1 (1%)
Serum LDH concentration		
Increased	72 (63%)	79 (69%)
Not increased	43 (37%)	35 (31%)
Presence of neurological symptoms at initial diagnosis	106 (92%)	105 (92%)
Histological result		
Total valid	113 (100%)	114 (100%)
DLBCL	111 (98%)	111 (97%)
Low-grade B-NHL	0	2 (2%)
Other*	2 (2%)	1 (1%)
CD20 positivity		
Total valid	114 (100%)	114 (100%)
Yes	113 (99%)	114 (100%)
No	1 (1%)	0
Missing	1 (1%)	0
Imaging		
Number of lesions		
Singular lesion	41 (36%)	49 (43%)
Multiple lesions	72 (64%)	65 (57%)
Missing	2 (2%)	0
Location		
Infratentorial manifestation	8 (7%)	5 (4%)
Supratentorial manifestation	83 (74%)	81 (72%)
Both	21 (19%)	27 (24%)
Missing	3 (3%)	1 (1%)
Meningeal involvement	5 (4%)	5 (4%)
Signs of leukoencephalopathy	4 (4%)	4 (4%)

(Table 1 continues in next column)

	R-DeVIC group (n=115)	HCT–ASCT group (n=114)
(Continued from previous column)		
Slit lamp result		
No signs of ocular infiltration	92 (80%)	90 (80%)
Ocular infiltration possible	4 (3%)	11 (10%)
Not analysed	19 (17%)	13 (11%)
CSF examination		
Cytology		
Positive	11 (10%)	5 (4%)
Suspicious	7 (6%)	12 (11%)
Negative	67 (58%)	75 (66%)
Other result	5 (4%)	1 (1%)
Not analysed	25 (22%)	21 (18%)
Protein concentration		
Increased	41 (35%)	43 (38%)
Not increased	35 (30%)	44 (38%)
Not analysed	39 (34%)	27 (23%)
Cumulative dexamethasone dose before treatment start, mg; median (IQR)	152·0 (84·0–240·0)	156·0 (72·0–272·0)

Data are n (%) unless otherwise indicated. There were no imbalances in potential confounders noted that could be due to chance or inconsistencies in randomisation. R-DeVIC=rituximab plus dexamethasone, etoposide, ifosfamide, and carboplatin. HCT–ASCT=high-dose chemotherapy followed by autologous stem-cell transplantation. ECOG=Eastern Cooperative Oncology Group. LDH=lactate dehydrogenase. DLBCL=diffuse large B-cell lymphoma. B-NHL=B-cell non-Hodgkin lymphoma. CSF=cerebrospinal fluid. *Other: R-DeVIC group: high-grade B-NHL, meningeosis lymphomatosa; HCT–ASCT group: Burkitt lymphoma.

Table 1: **Baseline patient characteristics**

randomisation, or if assessed as related to the study medication, until the end of the study.

Statistical analysis

The sample size was calculated based on the primary endpoint progression-free survival, assuming that the 3-year progression-free survival rate in the R-DeVIC group is approximately 50%. A hazard ratio (HR) of 1·8 of the R-DeVIC group versus the HCT–ASCT group was considered clinically relevant, corresponding to a progression-free survival rate after 3 years of 68% in the HCT–ASCT group.[14] To detect a difference between the R-DeVIC group and the HCT–ASCT group with a power of 80% at a two-sided significance level of α=0·05, a total of 92 progression-free survival-defining events was required. Assuming an exponential model for survival, an accrual period of 62 months, and an additional follow-up time of 2 years, we calculated a required enrolment of 210 patients. As follow-up might be incomplete for 5% of patients, 220 patients were to be randomised. Initially, it was anticipated that 10% of patients would not have a complete or partial response during induction. These figures were revised in an amendment, and subsequently, it was assumed that 15% of patients would not have a complete or partial response during induction treatment or would not be eligible for randomisation due to toxicity of the induction therapy or other reasons (about 20%), and thus would not reach randomisation.

The primary analysis was conducted in the full analysis set based on the intention-to-treat principle, excluding only patients with major violations of entry criteria (eg, absence of at least one measurable lesion, symptomatic coronary artery disease, cardiac arrhythmias uncontrolled with medication, myocardial infarction within the last

6 months) or patients not reaching complete response, unconfirmed complete response, or partial response immediately before randomisation. The per-protocol population was defined as a subgroup of the full analysis set, excluding those patients who did not receive randomised therapy. The safety analysis set contained all randomised patients who initiated therapy. Additionally, we analysed the data of all patients fulfilling entry criteria for registration in the trial and for whom induction therapy was started (registered patients).

The primary analysis of the progression-free survival endpoint was performed with the Cox proportional hazards model, containing the randomised treatment as the hypothesis variable adjusted for the stratification variable of response status. The treatment effect was described as the estimated HR with a two-sided 95% CI. The primary null hypothesis (equality of progression-free survival rates) was rejected if the value 1 was not contained in the two-sided 95% CI for the HR describing the relation between treatment groups. For statistical methods for secondary endpoints, refer to the statistical analysis plan in the appendix. All p values from analyses of secondary endpoints were interpreted in a descriptive sense. The analyses were performed using SAS version 9.4.

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

Between July 16, 2014, and Aug 31, 2019, 368 patients were enrolled, of whom 346 patients were registered within the trial and started induction therapy. Of these, 116 patients (34%) did not complete induction treatment; the main reasons were adverse events, serious adverse events, or treatment-related toxicity (n=59) and progressive or stable disease (n=36). During induction therapy, treatment was discontinued by 37 patients after the first cycle, 28 after the second cycle, 24 after the third cycle, and 27 after the fourth cycle; the median duration of therapy among patients who were not randomised was 34 days (IQR 6–65). Overall, 21 patients died within 100 days after registration, most commonly due to sepsis (n=9) or underlying lymphoma disease (n=6). Details on off-protocol treatments for the 116 patients who did not complete induction therapy are shown in the appendix (p 4). Subsequently, 230 (66%) of 346 patients were randomly assigned (1:1) to R-DeVIC or HCT–ASCT. As one patient in the HCT–ASCT group was excluded due to severe symptomatic cardiac disease, the full analysis set is based on 229 (66%) of 346 patients. 108 (94%) of the 115 patients in the R-DeVIC group and 111 (97%) of the 115 in the HCT–ASCT group started the allocated interventions. In the HCT–ASCT group, 100 patients received carmustine and 11 patients received busulfan. Thus, the safety analysis and per-protocol sets comprised

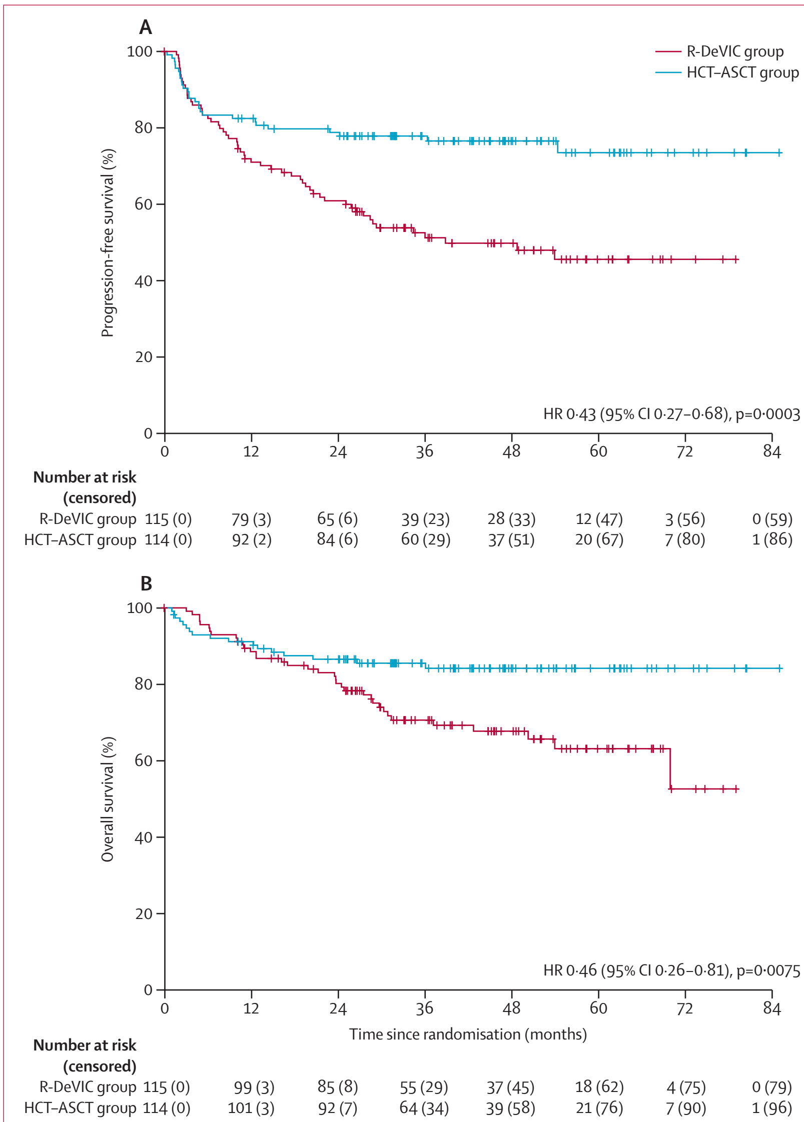

Figure 2: **Kaplan–Meier analysis of progression-free survival (A) and overall survival (B) over time for the full analysis set**

HRs were determined by Cox regression models adjusted for response status. HR=hazard ratio. R-DeVIC=rituximab plus dexamethasone, etoposide, ifosfamide, and carboplatin. HCT–ASCT=high-dose chemotherapy followed by autologous stem-cell transplantation.

219 (95%) of 230 randomised patients. Regarding consolidation treatment, seven (6%) of 115 patients in the R-DeVIC group and four (4%) of 115 patients in the HCT–ASCT group did not receive the allocated intervention (figure 1).

The distribution of baseline characteristics for the full analysis set was balanced across both treatment groups (table 1). The median time from the start of induction treatment to randomisation was 94 days (IQR 88–104). In the R-DeVIC group, 101 (94%) of 108 patients who at least started the allocated intervention received both treatment cycles with a median start of the second R-DeVIC cycle on day 28 of the first cycle (IQR 23–33). Dose intensity

	R-DeVIC group (n=115)	HCT–ASCT group (n=114)	Effect size (95% CI)
Rate of complete responses on day 60, n/N (% [95% CI])	75/115 (65% [56–74])	79/114 (69% [60–78])	OR 1·22 (0·68–2·18)
Overall survival			
Median (range), months	54 (44–not reached)	Not reached	..
3-year overall survival (95% CI)	71% (61–78)	86% (78–91)	HR 0·46 (0·26–0·81)
3-year cumulative incidence rate of relapse (95% CI)	45% (35–54)	19% (12–26)	SHR 0·40 (0·24–0·66)
EORTC QLQ-C30 median change in points (IQR)*			
Baseline to end of treatment	16·7 (–8·3 to 33·3)	8·3 (–16·7 to 25·0)	..
Baseline to 1-year follow-up	8·3 (–8·3 to 25·0)	16·7 (0 to 25·0)	..

Effect sizes are shown for the HCT–ASCT group compared with the R-DeVIC group. R-DeVIC=rituximab plus dexamethasone, etoposide, ifosfamide, and carboplatin. HCT–ASCT=high-dose chemotherapy followed by autologous stem-cell transplantation. OR=odds ratio. HR=hazard ratio. SHR=subdistribution hazard ratio. EORTC QLQ-C30=European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire. *Median change from baseline to end of treatment is reported for the 35/52 (67%) patients in the R-DeVIC group and 43/63 (68%) patients in the HCT–ASCT group, and from baseline to 1-year follow-up for the 23/52 (44%) patients in the R-DeVIC group and 27/63 (43%) patients in the HCT–ASCT group, who had available quality-of-life data at the respective timepoints (the denominators 52 [R-DeVIC group] and 63 [HCT–ASCT group] are the numbers of patients who had at least one baseline assessment at screening and did not experience progressive disease, relapse, or death).

Table 2: **Secondary efficacy endpoints for the full analysis set**

was comparable in both treatment groups and is displayed in the appendix (p 4). The median total number of infused $CD34^+$ cells in the HCT–ASCT group was $5 \cdot 3 \times 10^6$ per kg (IQR $3 \cdot 9$–$7 \cdot 9 \times 10^6$).

After a median follow-up of 45·3 months (IQR 31·7–58·3), 83 progression-free survival-defining events were reported in the full analysis set (56 in the R-DeVIC group *vs* 27 in the HCT–ASCT group; HR 0·43 [95% CI 0·27–0·68]; p=0·0003).

The progression-free survival probability over time from randomisation is displayed in figure 2 for the full analysis set and in the appendix (p 2) for the per-protocol set. In the full analysis set, the median progression-free survival in the R-DeVIC group was 39 months (95% CI 26–not reached), and not reached in the HCT–ASCT group. The progression-free survival rates at 12, 24, and 36 months were 71% (95% CI 62–78), 61% (51–69), and 51% (41–60), respectively, in the R-DeVIC group and 83% (74–88), 79% (70–85), and 78% (69–85), respectively, in the HCT–ASCT group. Progression-free survival for all patients who started induction (n=346) was calculated from registration. The progression-free survival rates at 12, 24, and 36 months were 66% (60–71), 58% (52–63), and 53% (47–58).

The median overall survival was not reached in either treatment group but was significantly prolonged following HCT–ASCT intervention (HR 0·46 [95% CI 0·26–0·81]; p=0·0075; table 2, figure 2B). Overall survival at 12, 36, and 60 months following randomisation was 89% (95% CI 81–93), 71% (61–78), and 63% (52–73) in the R-DeVIC group and 91% (84–95), 86% (78–91), and 84% (76–90) in the HCT–ASCT group for the full analysis set. 26 (23%) of the 115 patients assigned to the R-DeVIC group received further off-protocol treatment. Of these, 20 patients (77%) received HCT–ASCT. 12 (11%) of 114 patients who were assigned to the HCT–ASCT group received further off-protocol treatment. Details on applied off-protocol therapies for both treatment groups are provided in the appendix (p 4). For the full analysis set, complete response was documented for 75 (65%) of 115 patients in the R-DeVIC group and 79 (69%) of 114 in the HCT–ASCT group (HR 1·22 [95% CI 0·68–2·18]; table 2). Response rates on day 60 following randomisation are provided in the appendix (p 3).

Response duration is reported in terms of cumulative incidence rates of relapse. At 12, 24, and 36 months from complete response, unconfirmed complete response, or partial response documented at randomisation, these were 27% (95% CI 19–36), 37% (28–46), and 45% (35–54) in the R-DeVIC group, and 14% (8–21), 18% (11–25), and 19% (12–26) in the HCT–ASCT group (table 2).

Global health status at screening as per the EORTC QLQ-C30 questionnaire was available for 93 (81%) of 115 patients in the R-DeVIC group and 84 (74%) of 114 patients in the HCT–ASCT group, and was additionally collected at response assessments 2 and 3 and at year 1 during follow-up. Data on quality of life were analysed among patients with at least one baseline EORTC QLQ-C30 assessment at screening who did not experience progressive disease, relapse, or death (n=52 [R-DeVIC group] and n=63 [HCT–ASCT group]). The score range is 0–100 points, with higher values indicating a better global health status, as perceived by the patient. Median results showed a slightly more pronounced improvement over time in favour of the HCT–ASCT group given an overall high variability in the data (table 2, appendix p 3). Results of the Mini Mental State Examination, the detailed core neurocognitive test battery, and the EORTC QLQ-BN20 will be reported elsewhere.

The safety and tolerability of induction treatment were analysed in all 346 patients who received it. For the consolidation treatment, the safety and tolerability analyses were based on the safety analysis set comprising 219 randomised patients for whom consolidation treatment was started. Adverse events for the safety analysis set, grouped by MedDRA system organ class, are displayed in figure 3. The mean number of adverse events per patient was higher in the HCT–ASCT group (9·3 [SD 4·4] in the R-DeVIC group versus 14·6 [5·8] in the HCT–ASCT group). Detailed safety and tolerability data during induction and consolidation treatment are displayed in table 3. 200 patients (58%) reported at least one serious adverse event during induction treatment, with haematological toxicity and infectious complications being the most frequent. 18 patients (5%) experienced a fatal serious adverse event (the most frequent being infections and infestations [n=13] and respiratory disorders [n=3]) during induction treatment, of which 11 (61%) were considered at least possibly related to induction treatment.

During consolidation treatment, adverse events (of any grade) of febrile neutropenia, stomatitis, and diarrhoea were reported more frequently in the HCT–ASCT group than in the R-DeVIC group (61%, 79%, and 59% *vs* 14%, 9%, and 7%, respectively). Fatal serious adverse events following consolidation treatment occurred in two patients in the R-DeVIC group (both acute myeloid leukaemia) and in five patients in the HCT–ASCT group (infections and infestations [n=4], pulmonary embolism [n=1]). Besides the death from pulmonary embolism, the other six fatal serious adverse events during consolidation treatment were possibly related to treatment as judged by the investigators (table 3).

Discussion

Consolidative HCT–ASCT significantly improved both progression-free survival and overall survival compared with consolidative non-myeloablative chemotherapy with two cycles of R-DeVIC in patients with untreated PCNSL with at least partial response to MATRix induction, aged up to 70 years, in the MATRix/IELSG43 trial. Following general recommendations,[15] randomisation was performed as late as possible (ie, after induction treatment) to minimise selection bias that might be induced by early dropouts. Accordingly, the intention-to-treat population was well balanced due to randomisation after induction treatment and stratification according to response status. However, it is essential to note that the survival rates reported in this study pertain to patients who responded to induction therapy.

The rationale for HCT–ASCT in PCNSL is to deliver supra-conventional chemotherapy exposure, thereby enhancing drug diffusion across the blood–brain barrier and achieving CNS penetration not attainable with standard dosing. This therapeutic principle has been established in PCNSL since the late 1990s through various pilot and phase 2 studies.[5,7,9] In the current trial, progression-free survival and overall survival at 3 years were 78% and 86% after HCT–ASCT and 51% and 71% after R-DeVIC, respectively. Data on quality of life also show slight differences in favour of the HCT–ASCT group, with indications of a general increase over time, albeit against a backdrop of high variability. Although a more detailed analysis, additionally taking neurocognition into account, will be presented elsewhere, these findings suggest that at least a reduced quality of life in the HCT–ASCT group compared with the R-DeVIC group can be ruled out.

Two international, randomised, phase 2 trials previously compared HCT–ASCT with whole-brain radiotherapy of at least 36 Gy as consolidation in the first-line treatment of PCNSL, with both studies favouring HCT–ASCT regarding efficacy[9] and neurotoxicity.[7,13] Our results from the MATRix/IELSG43 trial—the largest randomised phase 3 study conducted in this disease—provide the highest level of evidence and further support HCT–ASCT as the standard consolidation approach in the first-line treatment of PCNSL.

Alternative non-myeloablative consolidation strategies have recently been tested, with the single-arm CALGB50202 study showing promising efficacy following high-dose etoposide and cytarabine after high-dose methotrexate-based induction therapy.[16] However, in the pivotal phase 2 CALGB 51101 trial, myeloablative therapy outperformed non-myeloablative treatment in terms of

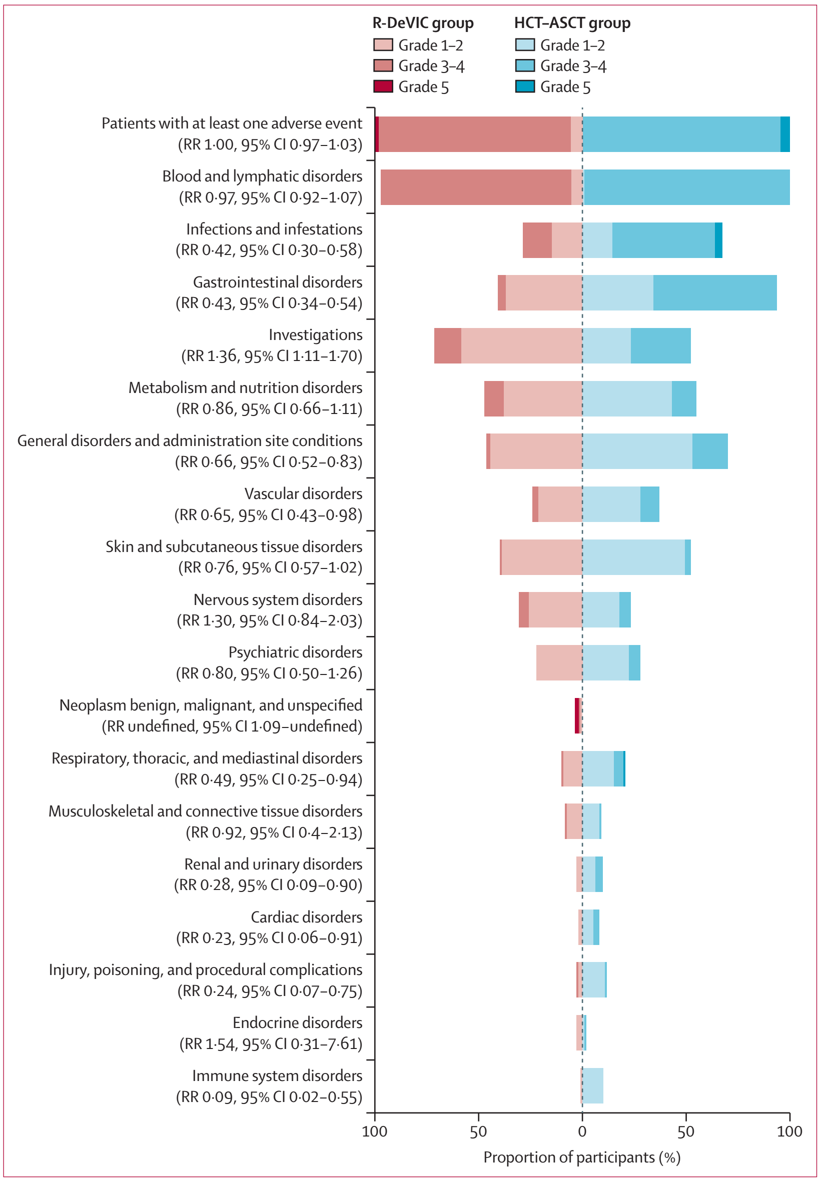

Figure 3: **Adverse events in the safety analysis set (n=219)**
Adverse events are grouped by MedDRA system organ class and graded according to Common Terminology Criteria for Adverse Events version 4.0. MedDRA system organ class totals are shown if the sum incidence of adverse events within one group was at least 10% (all grades) or if the sum incidence of adverse events being at least severe (grade ≥3) was at least 5%, or at least one fatal adverse event was reported within the group. MedDRA=Medical Dictionary for Regulatory Activities. RR=relative risk. R-DeVIC=rituximab plus dexamethasone, etoposide, ifosfamide, and carboplatin. HCT–ASCT=high-dose chemotherapy followed by autologous stem-cell transplantation.

2-year progression-free survival rates (73% *vs* 51%; p=0·02).[17] The CALGB 51101 trial randomly assigned participants to high-dose methotrexate-based remission induction followed by thiotepa-based HCT–ASCT or the same remission induction followed by non-myeloablative consolidation with etoposide and cytarabine.[17] However, the interpretation of these findings warrants caution. In CALGB 51101, progressive disease during induction occurred significantly more often in the non-myeloablative group than in the myeloablative group (24% *vs* 9%), which might have biased the primary comparison in favour of myeloablative consolidation. Notably, when efficacy is assessed from the start of consolidation rather than from randomisation, the apparent advantage of

	Induction (n=346)			Consolidation					
	All grades	Grade ≥3	Grade 5	R-DeVIC group (n=108)			HCT–ASCT group (n=111)		
				All grades	Grade ≥3	Grade 5	All grades	Grade ≥3	Grade 5
Patients with at least one adverse event	344 (99%)	340 (98%)	18 (5%)	108 (100%)	102 (94%)	2 (2%)	111 (100%)	111 (100%)	5 (5%)
Blood and lymphatic disorders	337 (97%)	332 (96%)	1 (<1%)	105 (97%)	99 (92%)	0	111 (100%)	110 (99%)	0
Thrombocytopenia	331 (96%)	328 (95%)	1 (<1%)	101 (94%)	90 (83%)	0	108 (97%)	106 (95%)	0
Anaemia	330 (95%)	299 (86%)	0	102 (94%)	75 (69%)	0	103 (93%)	82 (74%)	0
Leukopenia	313 (90%)	310 (90%)	0	89 (82%)	74 (69%)	0	101 (91%)	99 (89%)	0
Neutropenia	261 (75%)	252 (73%)	0	74 (69%)	61 (56%)	0	85 (77%)	82 (74%)	0
Febrile neutropenia	187 (54%)	185 (53%)	0	15 (14%)	15 (14%)	0	68 (61%)	67 (60%)	0
Infections and infestations	233 (67%)	179 (52%)	13 (4%)	31 (29%)	15 (14%)	0	75 (68%)	59 (53%)	4 (4%)
Device-related infection	55 (16%)	43 (12%)	0	3 (3%)	2 (2%)	0	23 (21%)	14 (13%)	0
Urinary tract infection	52 (15%)	32 (9%)	0	9 (8%)	7 (6%)	0	15 (14%)	11 (10%)	0
Pneumonia	49 (14%)	42 (12%)	4 (1%)	2 (2%)	2 (2%)	0	8 (7%)	8 (7%)	2 (2%)
Mucosal infection	42 (12%)	14 (4%)	0	2 (2%)	0	0	23 (21%)	13 (12%)	0
Skin infection	38 (11%)	15 (4%)	0	5 (5%)	0	0	4 (4%)	2 (2%)	0
Bronchitis	30 (9%)	22 (6%)	0	6 (6%)	1 (1%)	0	4 (4%)	4 (4%)	0
Sepsis	18 (5%)	18 (5%)	3 (1%)	0	0	0	3 (3%)	3 (3%)	1 (1%)
Abdominal infection	13 (4%)	13 (4%)	0	1 (1%)	1 (1%)	0	8 (7%)	8 (7%)	0
Neutropenic sepsis	7 (2%)	7 (2%)	1 (<1%)	0	0	0	3 (3%)	3 (3%)	1 (1%)
Pneumonia fungal	2 (1%)	1 (<1%)	0	0	0	0	1 (1%)	1 (1%)	1 (1%)
Septic shock	8 (2%)	8 (2%)	3 (1%)	0	0	0	3 (3%)	3 (3%)	0
Candida sepsis	1 (<1%)	1 (<1%)	1 (<1%)	0	0	0	0	0	0
Endocarditis staphylococcal	1 (<1%)	1 (<1%)	1 (<1%)	0	0	0	0	0	0
H1N1 influenza	1 (<1%)	1 (<1%)	1 (<1%)	0	0	0	0	0	0
Pseudomonal sepsis	2 (1%)	2 (1%)	1 (<1%)	0	0	0	0	0	0
Pulmonary sepsis	3 (1%)	3 (1%)	1 (<1%)	0	0	0	0	0	0
Gastrointestinal disorders	250 (72%)	50 (14%)	1 (<1%)	44 (41%)	4 (4%)	0	104 (94%)	66 (59%)	0
Stomatitis	117 (34%)	23 (7%)	0	10 (9%)	0	0	88 (79%)	60 (54%)	0
Nausea	113 (33%)	4 (1%)	0	23 (21%)	0	0	63 (57%)	13 (12%)	0
Constipation	74 (21%)	1 (<1%)	0	12 (11%)	0	0	9 (8%)	0	0
Diarrhoea	67 (19%)	11 (3%)	0	7 (7%)	3 (3%)	0	65 (59%)	12 (11%)	0
Abdominal pain	54 (16%)	7 (2%)	0	6 (6%)	0	0	23 (21%)	4 (4%)	0
Vomiting	50 (15%)	2 (1%)	0	9 (8%)	0	0	44 (40%)	1 (1%)	0
Investigations	313 (90%)	176 (51%)	0	77 (71%)	14 (13%)	0	58 (52%)	32 (29%)	0
Alanine aminotransferase increased	280 (81%)	143 (41%)	0	47 (44%)	2 (2%)	0	50 (45%)	5 (5%)	0
Gamma-glutamyltransferase increased	230 (66%)	62 (18%)	0	51 (48%)	11 (10%)	0	71 (64%)	27 (24%)	0
Aspartate aminotransferase increased	230 (66%)	67 (19%)	0	26 (24%)	0	0	48 (43%)	2 (2%)	0
Blood creatinine increased	134 (39%)	10 (3%)	0	21 (19%)	0	0	24 (22%)	2 (2%)	0
Blood bilirubin increased	119 (34%)	8 (2%)	0	8 (7%)	1 (1%)	0	18 (16%)	0	0

(Table 3 continues on next page)

	Induction (n=346)			Consolidation					
				R-DeVIC group (n=108)			HCT–ASCT group (n=111)		
	All grades	Grade ≥3	Grade 5	All grades	Grade ≥3	Grade 5	All grades	Grade ≥3	Grade 5
(Continued from previous page)									
Metabolism and nutrition disorders	236 (68%)	64 (18%)	0	51 (47%)	10 (9%)	0	61 (55%)	13 (12%)	0
Hyperglycaemia	152 (44%)	29 (8%)	0	29 (27%)	4 (4%)	0	25 (23%)	7 (6%)	0
Hyponatraemia	121 (35%)	20 (6%)	0	20 (19%)	4 (4%)	0	29 (26%)	0	0
Hyperkalaemia	72 (21%)	8 (2%)	0	11 (10%)	1 (1%)	0	10 (9%)	0	0
General disorders and administration site conditions	260 (75%)	59 (17%)	1 (<1%)	50 (46%)	2 (2%)	0	78 (70%)	19 (17%)	0
Pyrexia	165 (48%)	24 (7%)	1 (<1%)	10 (9%)	0	0	50 (45%)	4 (4%)	0
Pain	120 (35%)	19 (5%)	0	14 (13%)	0	0	25 (23%)	5 (5%)	0
Fatigue	110 (32%)	12 (3%)	0	25 (23%)	1 (1%)	0	37 (33%)	9 (8%)	0
Peripheral oedema	106 (31%)	4 (1%)	0	19 (18%)	1 (1%)	0	21 (19%)	1 (1%)	0
Vascular disorders	153 (44%)	55 (16%)	0	26 (24%)	3 (3%)	0	41 (37%)	10 (9%)	0
Hypertension	94 (27%)	31 (9%)	0	17 (16%)	2 (2%)	0	28 (25%)	8 (7%)	0
Hypotension	48 (14%)	13 (4%)	0	7 (6%)	0	0	15 (14%)	1 (1%)	0
Skin and subcutaneous tissue disorders	176 (51%)	3 (1%)	0	43 (40%)	1 (1%)	0	58 (52%)	3 (3%)	0
Alopecia	144 (42%)	0	0	38 (35%)	0	0	49 (44%)	0	0
Nervous system disorders	172 (50%)	42 (12%)	1 (<1%)	33 (31%)	5 (5%)	0	26 (23%)	6 (5%)	0
Headache	100 (29%)	5 (1%)	0	14 (13%)	0	0	14 (13%)	0	0
Psychiatric disorders	121 (35%)	35 (10%)	0	24 (22%)	0	0	31 (28%)	6 (5%)	0
Confusion state	83 (24%)	26 (8%)	0	9 (8%)	0	0	17 (15%)	4 (4%)	0
Depression	45 (13%)	5 (1%)	0	13 (12%)	0	0	10 (9%)	3 (3%)	0
Neoplasm benign, malignant, and unspecified	1 (<1%)	0	0	4 (4%)	3 (3%)	2 (2%)	0	0	0
Acute myeloid leukaemia	0	0	0	2 (2%)	2 (2%)	2 (2%)	0	0	0
Respiratory, thoracic, and mediastinal disorders	127 (37%)	31 (9%)	3 (1%)	11 (10%)	1 (1%)	0	23 (21%)	6 (5%)	1 (1%)
Epistaxis	68 (20%)	6 (2%)	0	6 (6%)	0	0	10 (9%)	0	0
Dyspnoea	55 (16%)	5 (1%)	0	4 (4%)	0	0	8 (7%)	2 (2%)	0
Pulmonary embolism	1 (<1%)	1 (<1%)	0	0	0	0	1 (1%)	1 (1%)	1 (1%)
Musculoskeletal and connective tissue disorders	119 (34%)	19 (5%)	0	9 (8%)	1 (1%)	0	10 (9%)	1 (1%)	0
Bone pain	85 (25%)	9 (3%)	0	4 (4%)	1 (1%)	0	5 (5%)	0	0
Muscular weakness	25 (7%)	7 (2%)	0	5 (5%)	0	0	4 (4%)	1 (1%)	0
Renal and urinary disorders	49 (14%)	8 (2%)	0	3 (3%)	0	0	11 (10%)	4 (4%)	0
Cardiac disorders	35 (10%)	13 (4%)	1 (<1%)	2 (2%)	0	0	9 (8%)	3 (3%)	0
Injury, poisoning, and procedural complications	35 (10%)	5 (1%)	0	3 (3%)	1 (1%)	0	13 (12%)	1 (1%)	0
Infusion-related reaction	16 (5%)	0	0	1 (1%)	0	0	12 (11%)	0	0
Endocrine disorders	33 (10%)	1 (<1%)	0	3 (3%)	0	0	2 (2%)	1 (1%)	0
Immune system disorder	32 (9%)	1 (<1%)	0	1 (1%)	0	0	11 (10%)	0	0
Hypersensitivity	32 (9%)	1 (<1%)	0	1 (1%)	0	0	11 (10%)	0	0

Data are n (%). Adverse events are shown if their incidence was at least 10% (all grades) or in case of being at least severe (grade ≥3) if their incidence was at least 5%. Fatal events were listed irrespective of frequency. System organ class totals are shown if their sum incidence was at least 10% (all grades) or in case of being at least severe (grade ≥3) was at least 5%, but no single adverse event's incidence was at least 10% or 5%. R-DeVIC=rituximab plus dexamethasone, etoposide, ifosfamide, and carboplatin. HCT–ASCT=high-dose chemotherapy followed by autologous stem-cell transplantation.

Table 3: **Adverse events**

myeloablative therapy might be no longer statistically significant. Although the different randomisation timepoints (upfront in the CALGB 51101 trial and before consolidation treatment in the MATRix/IELSG43 trial) add to the general limitations of inter-trial appraisal, the comparison of findings of the MATRix/IELSG43 trial with reported results of the IELSG32, PRECIS, and CALGB 51101 trials[4,9,17] is a useful tool to put the safety and efficacy of MATRix/IELSG43 strategies in context. Minor differences in eligibility criteria and randomisation strategy between IELSG32 and MATRix/IELSG43 should be considered. Nevertheless, toxicity and response rates after MATRix were similar in the IELSG32 and MATRix/IELSG43 trials, supporting the use of the same selection criteria across both trials and the robustness of the results. Importantly, actuarial progression-free survival and overall survival were similar in the HCT–ASCT group of the MATRix/IELSG43 trial and in the subgroup of patients treated with MATRix plus HCT–ASCT in the IELSG32 trial. Rates of toxicity, transplantation-related mortality, and serious adverse events after ASCT were similar in the IELSG32, MATRix/IELSG43, PRECIS, and CALGB 51101 trials, whereas comparable detailed reporting of these endpoints was not available for all other relevant studies, including CALGB50202. This should be considered when interpreting cross-trial safety comparisons.

Independent of consolidation-related toxic effects, morbidity during the induction phase is a critical determinant of outcome in patients with PCNSL. In MATRix/IELSG43, one-third of patients prematurely ended on-study treatment before randomisation, mainly due to progressive disease and treatment-related toxicity. Although the toxicity burden during MATRix induction was considerable, including treatment discontinuation due to toxicity and serious adverse events, it was in line with previously reported safety data for this regimen. These results are consistent with the safety reports of the MARTA and IELSG32 trials,[4,8] where around one-third of patients prematurely terminated high-dose methotrexate-based induction treatment, and highlight the importance of adequate supportive measures during the induction phase.

In all cases, there is a clear need to further optimise induction treatment strategies, aimed at better understanding patient-specific intolerability and identifying alternative pathways. The frequent infectious complications observed highlight the importance of proactive supportive care, including vigilant monitoring, early antimicrobial therapy, and prophylactic interventions, with implications for both routine clinical practice and the design of future studies.

Indeed, the ongoing phase 3 OptiMATe and PRIMA-CNS trials (DRKS00022768,[18] DRKS00024085[19]) are investigating a pretreatment phase with rituximab and high-dose methotrexate, aiming for a rapid improvement of performance status and corticosteroid tapering before initiation of more intensive induction treatment. Furthermore, recent phase 2 studies have shown encouraging response rates with the Bruton's tyrosine kinase inhibitors ibrutinib and tirabrutinib alone[20–22] or in combination[23] and the immunomodulatory agents lenalidomide and pomalidomide[24–26] in the relapsed or refractory setting of PCNSL. Consequently, these agents are currently being investigated in combination with high-dose methotrexate-based induction therapy in ongoing phase 2 trials (NCT04129710, NCT04446962). Additional emerging approaches to consolidation therapy in patients with PCNSL include chimeric antigen receptor (CAR) T-cell therapy.[27–31] Currently, retrospective case studies and small prospective phase 1–2 trials demonstrate the efficacy and tolerability of CAR T-cell therapy in relapsed or refractory PCNSL and secondary CNS involvement of B-cell lymphoma, but future prospective trials are warranted to explore the role of CAR T-cell therapy in patients with untreated PCNSL.

Limitations of our study should be considered. First, the survival rates reported here apply only to patients who responded to induction therapy, corresponding to 229 (66%) of the 346 patients. Accordingly, efficacy conclusions cannot be generalised to the full study cohort, and this highly intensive treatment approach is not universally applicable. Further limitations include the open-label design and the possibility of selection bias, since eligibility for HCT–ASCT, including assessment of cardiac, hepatic, and pulmonary fitness, was largely left to the investigators' discretion. In addition, despite the multinational design, the study population was restricted to western Europe, which might limit generalisability to other populations.

In conclusion, the MATRix/IELSG43 study, as the largest trial worldwide in the challenging setting of untreated PCNSL, provides important high-level evidence for the superiority of HCT–ASCT over non-myeloablative R-DeVIC as consolidation after high-dose methotrexate-based polychemotherapy in the patient group responsive to MATRix induction. International cooperative trials are presently exploring new strategies to further improve the tolerability and efficacy of treatment for this difficult-to-treat malignancy.

Contributors

GIl: conceptualisation, funding acquisition, writing—review and editing, supervision, resources, data interpretation. AJMF: conceptualisation, funding acquisition, resources, writing—review and editing. ES: conceptualisation, funding acquisition, writing—review and editing, data interpretation, resources. JW: writing—original draft, visualisation. MBi, PB, JH, SSt, AR, GE, TE, TW, BH, GL, MDr, RM, GK, CAS, UB, JP, MK, TH, SSc, TSL, LT, TP, HS, RN, GH, FPK, AKB, UK, EJP, MBP, FR, LO, LP, PLR: resources, writing—review and editing. MDe, MP: investigation. DG, MBa: methodology, formal analysis. AJ: formal analysis, visualisation. EV: funding acquisition, project administration, writing—review and editing. TC: investigation, resources. BK: investigation, writing—review and editing. MT: resources, investigation. HF, IS: verification. FS: methodology, investigation. EB: project administration. OG: project administration, verification. PvG, CH: investigation. GIh: data curation, formal analysis, writing—review and editing. JF: conceptualisation. EZ: conceptualisation. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication. The following authors directly accessed the data: GIl, ES, JW, AJMF, GIh, AJ, EV, and DG.

Declaration of interests

GIl receives research funding from Riemser Pharma and the German Federal Ministry of Research, Technology and Space (BMFTR); funding from Bristol Myers Squibb (BMS); honoraria for lecture fees from

Johnson & Johnson, Roche, Kite, and Amgen; travel support from Johnson & Johnson, Roche, and Gilead; and participates on advisory boards for BMS, Roche, and Kite. AJMF receives research funding from BMS, Pharmacyclics, Hutchison Medipharma, Amgen, Genmab, ADC Therapeutics, Gilead, and Novartis; receives consultation fees from Pfizer, Gilead, Novartis, Roche, PletixaPharm, Incyte, Genmab, AstraZeneca, and BMS; and reports receiving lecture fees from Adrienne, Gilead/Kite, Novartis, and Roche. JW reports receiving travel support from Lilly Pharma and is Co-speaker of the CNS working group of the German Lymphoma Alliance. PB reports receiving consulting fees from BMS, Roche, Miltenyi Biomedicine, Gilead/Kite, and Incyte; lecture fees from Takeda Oncology, BMS, Roche, Gilead/Kite, MSD, Miltenyi Biomedicine, Incyte, Sobi, and AbbVie; and receives support for attending meetings from Takeda Oncology, BMS, Roche, Gilead/Kite, MSD, Miltenyi Biomedicine, and Incyte. UB receives lecture fees from AbbVie, BeOne, and Incyte as well as travel support from AstraZeneca, BeOne, Gilead, Lilly, and Pierre Fabre; and participates on advisory boards for BeOne, AstraZeneca, Janssen, and Pierre Fabre. DG receives a grant from Eva Mayr-Stihl Foundation. JH receives lecture fees from AbbVie, AstraZeneca, and Novartis as well as travel support from Gilead and participates on advisory boards for BMS, Janssen, Jazz, and MSD. BK reports receiving consulting fees from Roche, Incyte, and Astellas and travel support from Takeda. GK receives lecture fees from AbbVie, Amgen, Biotest, BMS/Celgene, Eurocept, Janssen/Cilag, Kite/Gilead, Novartis, Medac, MSD, Pfizer, Sanofi, and Takeda as well as travel support from Medac, Neovii, and Sanofi. FPK reports receiving travel support from BeiGene, Lilly, and Janssen, and participates on an advisory board for Roche. PLR receives honoraria from Roche. LP reports receiving lecture fees from Jazz and Novartis, as well as travel support from Johnson & Johnson and Medac, and participates on an advisory board for Alexion. AR reports lecture honoraria and consultation fees from Alexion, Amgen, Apellis, Biocryst, Bioverativ, Kira, Novartis, Roche, Recordati, Sanofi, and Sobi. GL receives grants from AbbVie, Sobi, Gilead, Roche and Incyte; consulting fees from Roche, Gilead, GSK, Novartis, AstraZeneca, BMS, AbbVie, Incyte, ADC Therapeutics, PentixaPharm, Sobi, Immagene, Hexal, Lilly, MSD, BeiGene/BeOne, and Pierre Fabre; lecture fees from Roche, Gilead, AstraZeneca, AbbVie, Sobi, Incyte, BMS, Hexal, MSD, and Pierre Fabre; travel support from Roche, AbbVie, Sobi, BeiGene/BeOne, Gilead, AstraZeneca, BMS, and Incyte; participates on advisory boards for BMS; and is president of the German Lymphoma Alliance. MDr reports receiving research funding from AbbVie, Gilead/Kite, Janssen, Lilly, and Roche; receives consultancy fees from AbbVie, Gilead/Kite, AvenCell, Janssen, Roche, AstraZeneca, Lilly, BeOne, BMS, Genmab, Incyte, Sobi, and Novartis; received honoraria from AbbVie, AstraZeneca, BeOne, BMS, Gilead/Kite, Janssen, Lilly, Roche, Novartis, and Sobi; and is a member of the American Society of Hematology, American Society of Clinical Oncology, Deutsche Gesellschaft für Hämatologie und Medizinische Onkologie, European Hematology Association, European Society for Medical Oncology, and German Lymphoma Alliance. TSL reports receiving lecture fees from Bristol Myers Squibb, and Gilead/Kite, as well as payment for expert testimony from Johnson & Johnson; and participates on advisory boards for AstraZeneca, Gilead/Kite, BMS, and Roche. HS reports receiving consulting fees and/or lecture fees from AbbVie, Amgen, AstraZeneca, BMS/Celgene, GSK, Janssen, Oncopeptides, Pfizer, Regeneron, Roche, Sanofi, and Menarini Stemline; and travel support from Amgen, BMS/Celgene, Janssen, Pfizer, Sanofi, and Menarini Stemline. GH received honoraria and/or consulting fees from AbbVie, ADC Therapeutics, AstraZeneca, BeOne, Gilead/Kite, Incyte, Janssen, Novartis, BeiGene, BMS, Lilly, MSD, Sobi, and Roche; and receives additional travel support from Sobi and Janssen and research funding from Gilead/Kite, Janssen, AbbVie, and Roche. LT receives travel support from Johnson & Johnson and is speaker of the Burkitt Lymphoma sub-committee of the German Lymphoma Alliance; and holds a patent for progranulin as a marker for autoimmune diseases. EZ reports receiving grants from AstraZeneca, Celgene/BMS, BeiGene/BeOne, Incyte, Janssen, and Roche as well as consulting fees from Incyte, lecture fees from AbbVie and Roche, and travel support from AbbVie, AstraZeneca, BeiGene/BeOne, and Gilead; and participates on advisory boards for AbbVie, AstraZeneca, BMS, Eli/Lilly, Incyte, Ipsen, and Sobi. ES receives study support from Riemser Pharma and the BMFTR and a grant from BMS; participates on advisory boards for BMS and SERB SAS Pharmaceuticals; and is speaker of the CNS lymphoma section of the German Lymphoma Alliance. TW reports receiving lecture fees from AbbVie, AstraZeneca, BeOne, Gilead/Kite, Ideogen, Roche, Takeda, Lilly, and Johnson & Johnson; payment for expert testimony from SecuraBio, Takeda, and Pentixapharm; research support from Clinigen, Esteve, Ideogen, PentixaPharm, Riemser, and Takeda; and received travel support from AbbVie, AstraZeneca, Gilead/Kite, Johnson & Johnson, Roche, Takeda, and BeOne. All other authors declare no competing interests.

Data sharing

A data sharing plan was included in the original trial's registration. The data sharing plan did not change after registration. Briefly, individual participant data will not be shared. The clinical trial protocol has been published as an open-access publication (https://doi.org/10.1186/s12885-016-2311-4). Aggregated participant data are publicly available on the EudraCT registry (https://www.clinicaltrialsregister.eu/ctr-search/search?query=eudract_number:2012-000620-17) and as part of the Article publication.

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